JONATHAN MENDOZA

Mechanical Engineer, EIT
linkedin.com/in/jm139 • mendozajonathan053@gmail.com • (213) 239-4046 • jsm139.github.io/portfolio

PROFESSIONAL SUMMARY

Mechanical Engineer with 5+ years of experience in mechanical product development, CNC machine systems, prototyping, failure analysis, and engineering automation. Experienced in SOLIDWORKS design, assemblies, engineering analysis, manufacturing support, and cross-functional product development from design through release.

EXPERIENCE

TRAK Machine Tools / Southwestern Industries, Rancho Dominguez, CA
Mechanical Engineer — Feb 2021 – Present

- Designed and modified 100+ mechanical parts and assemblies for CNC machines using SOLIDWORKS, supporting prototyping, manufacturing, and production release.
- Created and released 100+ engineering drawings using GD&T and ASME Y14.5 to define tolerances and manufacturing requirements.
- Managed 50+ ECN/NPI activities using SolidWorks Manage/PLM, coordinating changes across engineering, manufacturing, suppliers, and quality.
- Managed BOMs for TRAK's current CNC product portfolio, maintaining product structures and engineering documentation.
- Conducted failure analysis using vibration measurements to isolate mechanical issues and compare frequencies against ISO 10816. Documented findings, recommended solutions, and verified corrective actions.
- Supported hands-on development through 3D printing, rapid prototyping, inspection, and CNC manufacturing while working with engineers, machinists, suppliers, and quality personnel.
- Developed 10+ SolidWorks API/VBA macros to automate repetitive drafting and modeling tasks.

Maxwell Vehicles, Los Angeles, CA (Remote)
Mechanical Engineering Intern — Aug 2020 – Nov 2020

- Designed shipping-package components and assemblies using SOLIDWORKS for Maxwell's electric ProMaster (ePRO).
- Created build drawings and organized Bills of Materials to support technician assembly and manufacturing.
- Worked independently while communicating project progress, questions, and feedback to the project supervisor.

TECHNICAL SKILLS

Mechanical Design: SOLIDWORKS, Assemblies, Engineering Drawings, GD&T, ASME Y14.5, DFM/DFA

Analysis & Testing: SolidWorks Simulation, FEA, Failure Analysis, Root Cause Analysis, Vibration Measurement, ISO 10816, Data Acquisition, Inspection

Manufacturing & Prototyping: CNC Machining, 3D Printing, Rapid Prototyping, Fabrication, Production Support, NPI, ECN

Engineering Tools: SOLIDWORKS PDM/Manage, AutoCAD, Inventor, DraftSight, SolidWorks API, VBA, Python, MATLAB

Languages: English (Native), Spanish (Fluent), Portuguese (Fluent)

EDUCATION

California State University, Los Angeles
Bachelor of Science in Mechanical Engineering • Minor in Computer Science — Sept 2015 – May 2020

CERTIFICATIONS

- Engineer in Training (EIT), NCEES, Issued Feb 2020
- Certified SOLIDWORKS Professional (CSWP), Dassault Systèmes, Issued Nov 2020

SELECTED PROJECT

Design and Fabrication of a Crash Sled System — Jan 2021 – May 2021
Senior Capstone Project, California State University, Los Angeles

- Led a team of 4 students in the mechanical design and fabrication of a crash sled system for a university research project.
- Developed SOLIDWORKS models and manufacturing drawings and performed stress analysis on key components to evaluate structural durability.